Strategy Brief #9

Application of SSA for Land Subsidence Time Series analysis

Demonstrating how Singular Spectrum Analysis (SSA), by decomposing complex, noisy subsidence data into trend, periodic, and noise components, overcomes the limitations of traditional methods to deliver more accurate long-term risk detection and stronger scientific foundations for infrastructure and policy planning.

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Introduction

Understanding land subsidence dynamics requires robust analysis of complex and noisy time series data. Advanced methods such as **Singular Spectrum Analysis (SSA)** offer powerful tools to extract meaningful patterns and trends.

Key Challenge

Traditional analytical approaches often fail to capture **non-linear dynamics and hidden signals** in subsidence data, limiting the accuracy of risk assessments.

Strategic Insights

- SSA allows decomposition of time series into **trend, periodic, and noise components**.
- It improves the detection of **long-term subsidence trends and anomalies.**
- Enhances the reliability of **forecasting models**.
- Can be applied to **data-scarce and noisy environments.**

Strategic Implications

- Improved data interpretation supports **better-informed decision-making**.
- Enables **early identification of critical subsidence trends**.
- Strengthens the scientific basis for **policy and infrastructure planning.**

Recommendations

- Integrate SSA into **standard analytical workflows for subsidence monitoring.**
- Train technical teams in **advanced time series analysis methods.**
- Combine SSA with **machine learning and geospatial analysis.**
- Promote adoption in **research and operational monitoring systems.**

Value Proposition

Advanced analytical methods like SSA enhance the **accuracy, reliability, and strategic value of environmental data analysis**.